

accuracy elec. structure

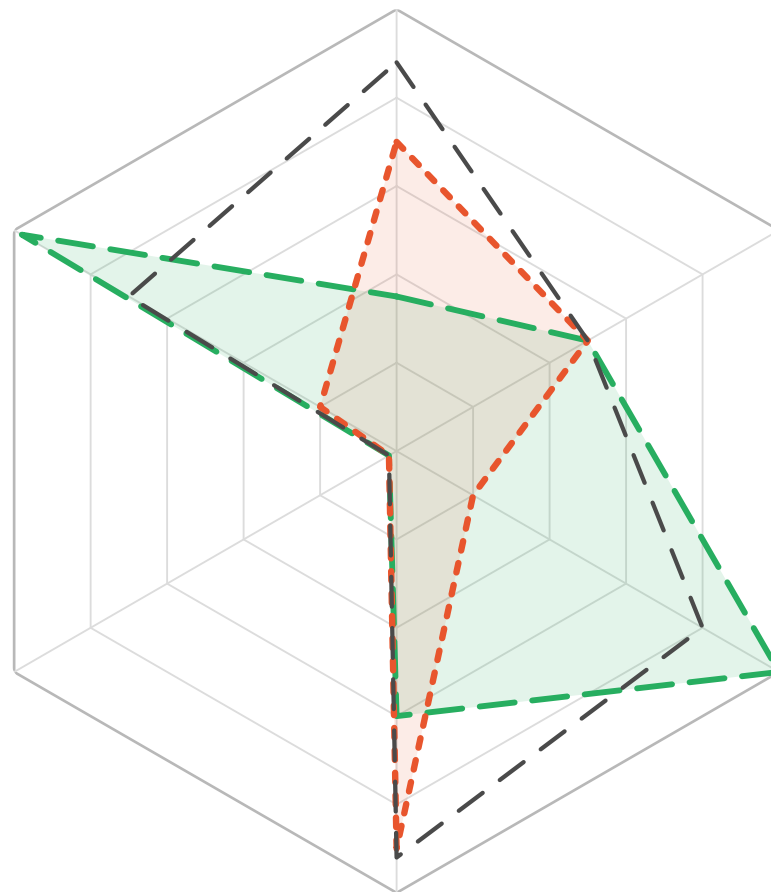
force field $\rightarrow$ multireference

degrees of freedom

2 – 3 $\rightarrow$ 10^5

accuracy nuclei dynamics

classical $\rightarrow$ semi-classical $\rightarrow$ quantum



light-matter interaction

implicit $\rightarrow$ explicit field

electron dynamics

no $\rightarrow$ yes

time scale of the process

fs $\rightarrow$ ns

— — classical MD

- - - AIMD (DFT)

— — MLIP